

Generalized linear models in R

Jeff Leek

- Dependencies
- Logistic regression
 - Load the data
 - A single logistic regression
 - Fit many glms at once
- Poisson/negative binomial regression
 - Download the data
 - Transform the data
 - A single Poisson regression
 - Multiple negative binomial regressions
- More information
- Session information

Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(snpStats)
library(broom)
library(MASS)
library(DESeq2)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools", "broom", "MASS"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase", "snpStats", "DESeq2"))
```

Logistic regression

Load the data

Here we use example SNP data from a case-control genome-wide association study Load an example data set and take a smaller subset of samples for computational efficiency

```
data(for.exercise)
use <- seq(1, ncol(snpStats), 10)
sub.10 <- snpStats[,use]
```

Calculate the PCs

```
xxmat <- xxt(sub.10, correct.for.missing=FALSE)
evv <- eigen(xxmat, symmetric=TRUE)
pcs <- evv$vectors[,1:5]
```

A single logistic regression

First we do an unadjusted logistic regression assuming an additive model. The coefficient is the change in log-odds for a one unit decrease (because homozygous major allele is coded 1) in the number of copies of the minor allele.

```
snpdata = sub.10@.Data
status = subject.support$cc
snp1 = as.numeric(snpdata[,1])
snp1[snp1==0] = NA
glm1 = glm(status ~ snp1, family="binomial")
tidy(glm1)
```

```
##           term      estimate std.error  statistic    p.value
## 1 (Intercept)  0.1122026  0.2322000   0.4832155  0.6289427
## 2          snp1 -0.1010806  0.2012159  -0.5023489  0.6154221
```

We can also easily code in other models. For example suppose we want to code a dominant model (so only an association of risk with the two copies of the common allele, now the coefficient on `snp1_dom` is the increase in log odds associated with two copies of the major allele).

```
snp1_dom = (snp1 == 1)
glm1_dom = glm(status ~ snp1_dom, family="binomial")
tidy(glm1_dom)
```

```
##           term      estimate std.error  statistic    p.value
## 1 (Intercept) -0.1112256  0.1927477  -0.5770531  0.5639036
## 2 snp1_domTRUE  0.1248313  0.2041740   0.6113966  0.5409371
```

```
tidy(glm1)
```

```
##           term      estimate std.error  statistic    p.value
## 1 (Intercept)  0.1122026  0.2322000   0.4832155  0.6289427
## 2          snp1 -0.1010806  0.2012159  -0.5023489  0.6154221
```

We can also easily adjust for other variables.

```
glm2 = glm(status ~ snp1 + pcs[,1:5], family="binomial")
tidy(glm2)
```

```
##           term      estimate std.error  statistic    p.value
## 1 (Intercept)  0.08993531 0.2341663   0.38406597 0.700929553
## 2          snp1 -0.08072045 0.2029343  -0.39776640 0.690802386
## 3 pcs[, 1:5]1  5.11070256 2.0261844   2.52232842 0.011658081
## 4 pcs[, 1:5]2  0.16534907 2.0365104   0.08119235 0.935288981
## 5 pcs[, 1:5]3  2.40608659 2.0434133   1.17748405 0.239002361
## 6 pcs[, 1:5]4 -5.39830830 2.0466337  -2.63765245 0.008348209
## 7 pcs[, 1:5]5  0.58408419 2.0355288   0.28694470 0.774154663
```

Fit many glms at once

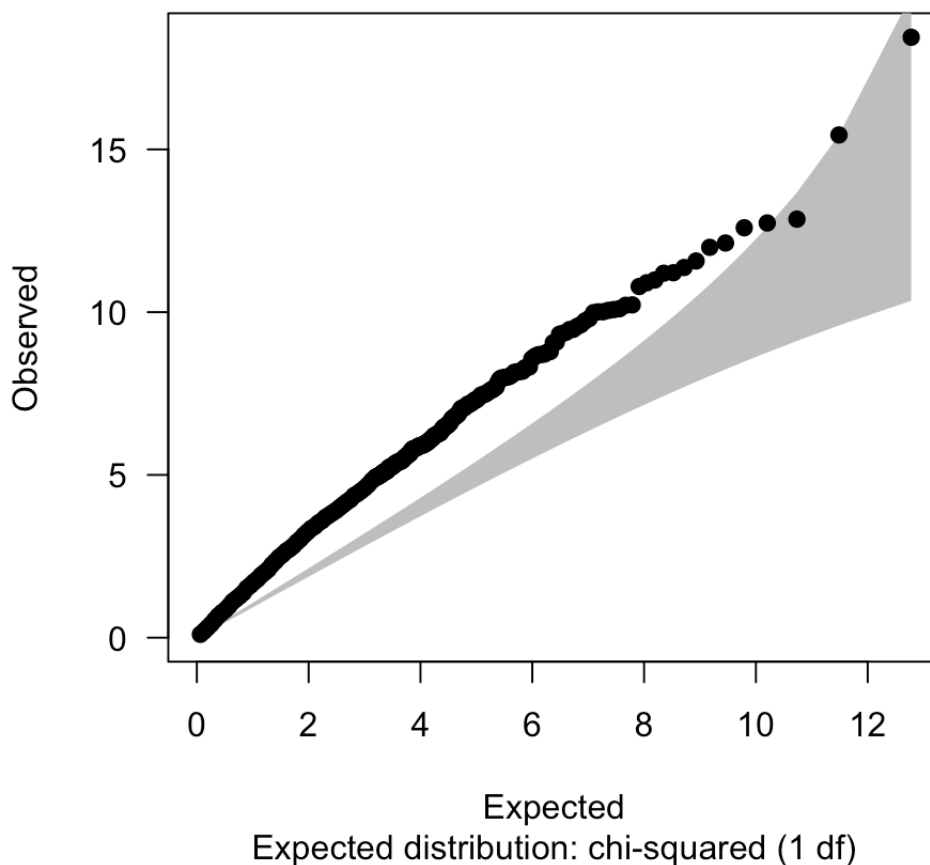
For logistic regression modeling of many SNPs at once we can use the `snps.rhs.tests` function which computes an asymptotic chi-squared statistic. This isn't quite the same thing as the F-statistics we have been calculating but can be used in the same way for significance calculations.

```
glm_all = snp.rhs.tests(status ~ 1, snp.data=sub.10)
slotNames(glm_all)
```

```
## [1] "snp.names" "var.names" "chisq"    "df"      "N"
```

```
qq.chisq(chi.squared(glm_all), df=1)
```

QQ plot

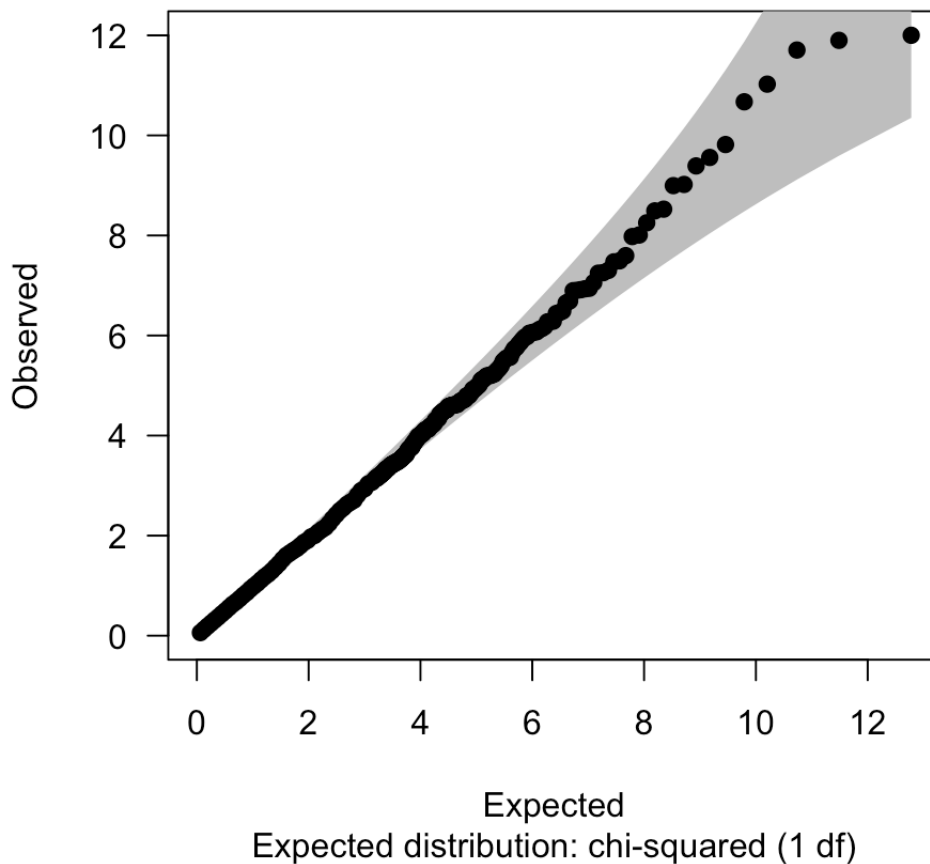


##	N	omitted	lambda
##	2851.000000	0.000000	1.657571

We can also adjust for variables like principal components

```
glm_all_adj = snp.rhs.tests(status ~ pcs,snp.data=sub.10)
qq.chisq(chi.squared(glm_all_adj),df=1)
```

QQ plot



##	N	omitted	lambda
##	2851.000000	0.000000	0.9770849

Poisson/negative binomial regression

Download the data

Here we are going to use some data from the paper Evaluating gene expression in C57BL/6J and DBA/2J mouse striatum using RNA-Seq and microarrays. (<http://www.ncbi.nlm.nih.gov/pubmed?term=21455293>) that is a comparative RNA-seq analysis of different mouse strains.

```

con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata=pData(bot)
edata=as.matrix(exprs(bot))
fdata = fData(bot)
ls()

```

```

## [1] "bot"           "bottomly.eset" "con"
## [4] "edata"         "evv"           "fdata"
## [7] "glm_all"       "glm_all_adj"   "glm1"
## [10] "glm1_dom"      "glm2"          "pcs"
## [13] "pdata"         "snp.support"   "snp1"
## [16] "snp1_dom"      "snpdata"       "snps.10"
## [19] "status"        "sub.10"        "subject.support"
## [22] "tropical"      "use"           "xxmat"

```

Transform the data

Here we remove lowly expressed genes but we will leave them as counts

```
edata = edata[rowMeans(edata) > 10, ]
```

A single Poisson regression

The coefficient in this case is the increase in the log number of counts comparing one strain to the other. If you exponentiate the coefficients then $\exp(\text{intercept})$ is expected count for C57BL/6J and $\exp(\text{intercept}) \exp(\text{strainDBA}/2J)$ is the expected count for the strain DBA/2J.

```

glm3 = glm(edata[1, ] ~ pdata$strain,family="poisson")
tidy(glm3)

```

```

##           term      estimate std.error  statistic    p.value
## 1 (Intercept) 6.23988591 0.01396450 446.839073 0.000000000
## 2 pdata$strainDBA/2J 0.04982967 0.01907011   2.612972 0.008975864

```

You can also fit a negative binomial regression one at a time in R with:

```

glm.nb1 = glm.nb(edata[1, ] ~ pdata$strain)
tidy(glm.nb1)

```

```

##           term      estimate std.error  statistic    p.value
## 1 (Intercept) 6.23988591 0.1205228 51.7735102 0.0000000
## 2 pdata$strainDBA/2J 0.04982967 0.1665002   0.2992768 0.7647288

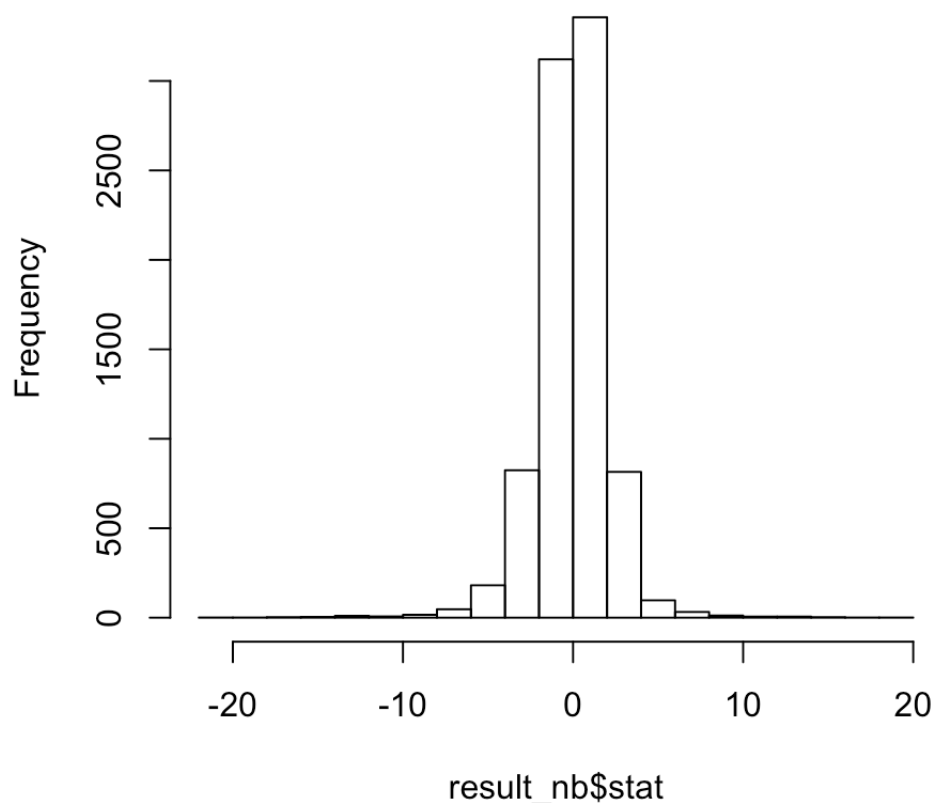
```

Multiple negative binomial regressions

We can use the `DESeq2` package to perform many (moderated) negative binomial regressions at once. We first need to create a DESeq data set.

```
de = DESeqDataSetFromMatrix(edata, pdata, ~strain)
glm_all_nb = DESeq(de)
result_nb = results(glm_all_nb)
hist(result_nb$stat)
```

Histogram of result_nb\$stat



More information

You can find a lot more information on this model fitting strategy in:

- These lecture notes on glms in R (<http://data.princeton.edu/R/glms.html>)
- The snpStats vignettes (<http://bioconductor.org/packages/release/bioc/html/snpStats.html>)
- The RNA-seq workflow (<http://www.bioconductor.org/help/workflows/rnaseqGene/>) for Bioconductor

Session information

Here is the session information

```
devtools::session_info()
```

```
## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12
## GO.db 3.1.2 2015-09-06
```

##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19
##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25


```
## snm                1.16.0      2015-04-17
## snpStats           * 1.18.0      2015-04-17
## stringi            0.5-5        2015-06-29
## stringr            1.0.0        2015-04-30
## survival           * 2.38-3      2015-07-02
## sva                 * 3.14.0      2015-04-17
## tidyr              0.2.0        2014-12-05
## UsingR             * 2.0-5        2015-08-06
## whisker            0.3-2        2013-04-28
## XML                3.98-1.3      2015-06-30
## xml2               0.1.2        2015-09-01
## xtable             1.7-4        2014-09-12
## XVector            0.8.0        2015-04-17
## yaml              2.1.13        2014-06-12
## zlibbioc           1.14.0        2015-04-17
## source
## CRAN (R 3.2.0)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## Github (jdstorey/edge@a1947b5)
## CRAN (R 3.2.2)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
```

```
## Bioconductor
## Bioconductor
## local
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafrazee/RSkittleBrewer@0a96a20)
```

```
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.0)
## Bioconductor
```

It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.